

Exploratory Data Analysis and Visualization of Retail Transactions

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Abstract—This study explores customer purchasing habits, product performance, and temporal sales trends using a 30-day retail dataset of 292 transactions from 20 users. Through Exploratory Data Analysis (EDA) and 15 multi-visualizations, we map the connections between user engagement and revenue generation. Our findings reveal highly variable spending behaviors, a strong correlation between basket size and transaction value, and significant revenue concentration driven by specific items and peak days (Gini Coefficient: 0.464). Ultimately, this research highlights how data-driven visualization techniques can directly enhance practical retail decision-making, inventory optimization, and customer segmentation.

I. INTRODUCTION

Understanding consumer behavior is essential for modern retail optimization. While traditional metrics like gross revenue obscure individual habits and temporal trends, data-driven visual methods can uncover deep insights hidden within raw transaction logs.

To illustrate this, our research applies Exploratory Data Analysis (EDA) to a 30-day retail dataset. We investigate three core dimensions: User Analysis (engagement), Item Analysis (demand and affinities), and Transaction Analysis (trends and inequality), demonstrating how visual analytics empowers strategic decision-making.

II. METHODOLOGY

A. Dataset Description

The dataset consists of a 30-day simulated retail log. During the parsing and preprocessing phase, the following overall summary statistics were extracted and utilized as the baseline for all subsequent metrics:

- **Total Users:** 20
- **Total Transactions:** 292
- **Total Unique Items:** 26
- **Total Revenue:** Rs. 574,262.07
- **Average Transaction Value:** Rs. 1,966.65
- **Average Basket Size:** 8.23 items
- **Avg Transactions/User:** 14.60

B. Data Preprocessing

- 1) Parsed raw text logs to extract User IDs, Item Identifiers (A-Z), and Prices.
- 2) Aggregated daily transactions to map temporal purchasing rhythms.
- 3) Normalized monetary values to Indian Rupees (Rs.) and verified data integrity for missing inputs.

C. Exploratory Data Analysis (EDA)

- 1) Visualized user frequency and revenue distributions using bar charts and small multiple area charts.
- 2) Analyzed item co-occurrence (Market Basket Analysis) to find frequently bundled products.
- 3) Calculated statistical metrics, including Pearson correlation coefficients, skewness, standard deviations, and the Gini Coefficient for revenue inequality.

III. RESULTS

The exploratory analysis showed notable relationships between various input features and the target variables across users, items, and transactions. The resulting visualizations have been grouped into their respective analytical categories.

A. User Analysis

Initial user analysis reveals significant variance in customer engagement. Evaluating the transaction frequency across all 20 users identifies the most engaged shoppers. Users 15 and 19 exhibit the highest engagement with 20 transactions each, whereas User 12 records the lowest with only 9 transactions. This demonstrates a wide spectrum of store visitation frequency among the customer base. Recognizing these high-frequency users allows retailers to focus retention and loyalty programs directly on the individuals driving the majority of foot traffic.

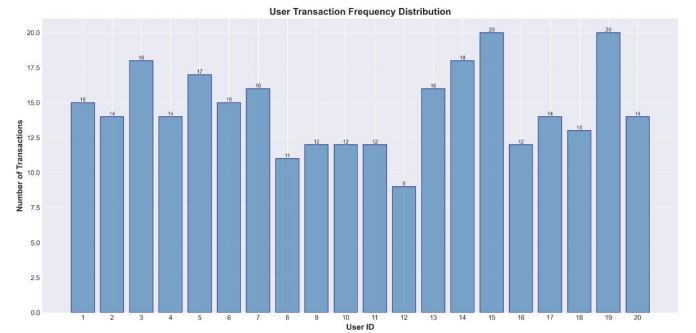


Fig. 1. User Transaction Frequency Distribution

Ranking the users by their total monetary spending reveals the true financial drivers of the store. User 19 emerges as the most valuable customer, generating Rs. 38,522.57 in total revenue, which correlates with their high visitation frequency. In contrast, User 12 generated the least revenue at Rs. 15,171.94.

This highlights the necessity of segmenting users based on their lifetime value rather than mere visitation counts, ensuring high-value clients receive appropriate VIP tiering.

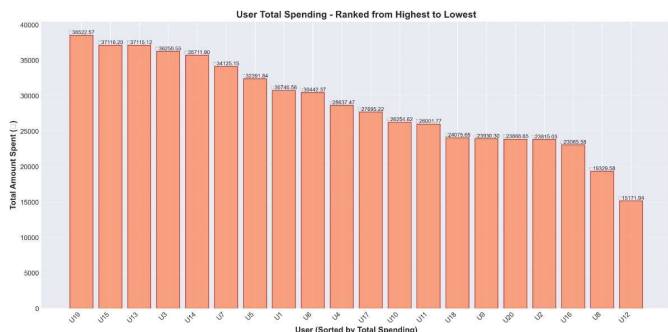


Fig. 2. User Total Spending Ranked from Highest to Lowest

To observe individual shopping habits dynamically, a Small Multiples Grid isolates the 30-day spending trends for each user on a shared scale. The visualizations show highly erratic, spike-based purchasing behavior (e.g., User 3 and User 14) rather than smooth, consistent daily spending. This suggests that customers engage in large, periodic "stock-up" trips rather than minor daily purchases. The shared Y-axis allows for immediate visual comparison, revealing that some users' maximum spending spikes completely dwarf the day-to-day spending of the rest of the cohort.

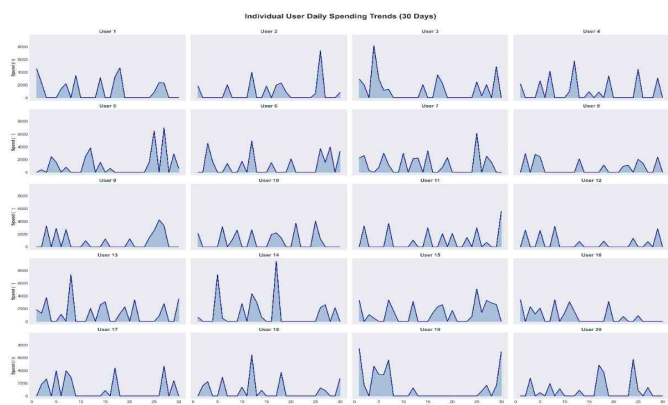


Fig. 3. Individual User Daily Spending Trends across 30 Days (Small Multiples)

Analyzing the purchasing rhythm of the customer base involves calculating the days elapsed between consecutive transactions. The distribution reveals a median of 1.00 day and a mean of 1.99 days, indicating a highly loyal customer base that returns frequently. However, severe outliers (up to 14 days) exist, highlighting a subset of customers with extended purchase cycles. By identifying these specific temporal gaps, marketing teams can accurately time re-engagement emails to prevent churn.

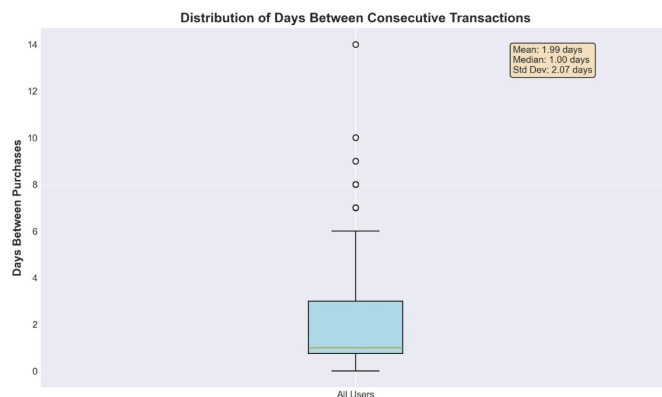


Fig. 4. Distribution of Days Between Consecutive Transactions

Categorizing each user's purchases into unique versus repeated items provides insight into brand loyalty and catalog exploration. A significant portion of every user's total basket consists of repeated items, indicating that users strongly prefer to buy familiar products routinely rather than constantly exploring new items in the catalog. Because user habits appear highly rigid, introducing new inventory to this customer base may require aggressive discounting or bundling.

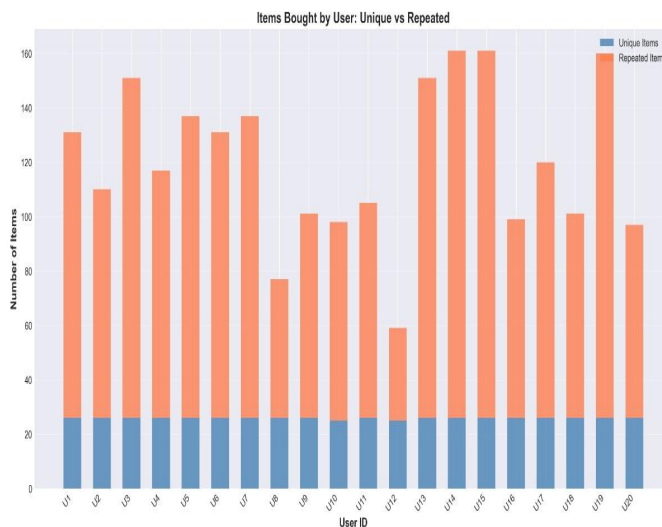


Fig. 5. Items Bought by User: Unique vs Repeated

B. Item Analysis

Analysis of product performance demonstrates that transaction volume does not necessarily mirror financial value. Evaluating the pure volume of items sold shows that item 'P' is the most frequently purchased (104 times), closely followed by item 'B' (101 times). The least frequent item, 'N', was still purchased 85 times, indicating a relatively flat frequency distribution across the catalog. This tight frequency band suggests that there is no "dead stock" taking up space; all items show steady movement.

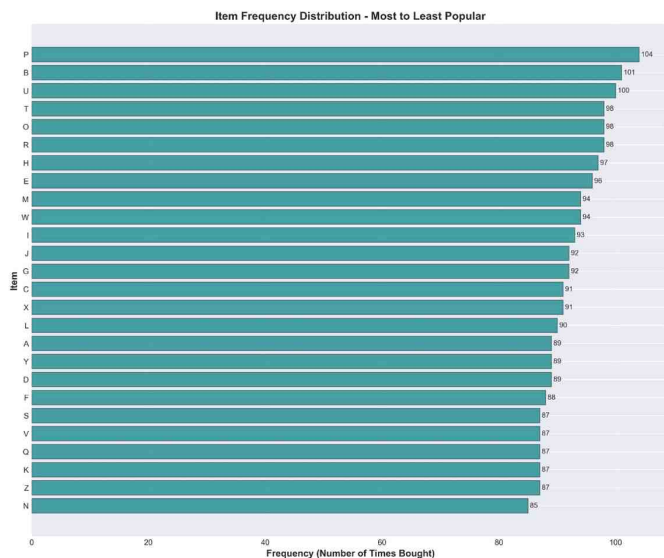


Fig. 6. Item Frequency Distribution - Most to Least Popular

Conversely, evaluating the total revenue generated by each item reveals stark contrasts. Item ‘V’ is the top revenue generator, contributing over Rs. 40,000, significantly outperforming the most frequently bought item, ‘P’, which generated approximately Rs. 25,000. This disparity underscores the importance of pricing weight in overall revenue generation. Low-volume, high-margin items require drastically different supply chain logistics compared to fast-moving staples.

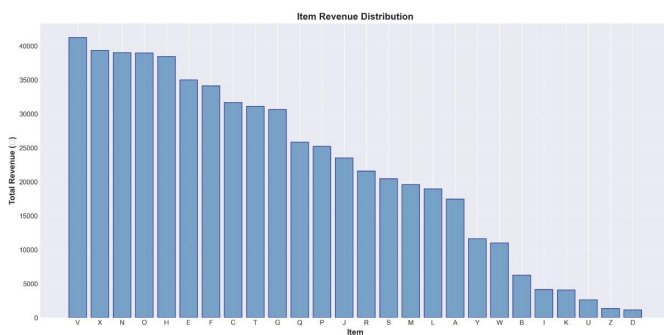


Fig. 7. Item Revenue Distribution

To inform cross-selling strategies, a Market Basket Analysis was performed targeting the highest-volume item, ‘P’. The analysis reveals that items ‘I’ (43 co-occurrences), ‘G’ (42), and ‘R’ (42) possess the strongest affinities with ‘P’. These pairings present prime opportunities for strategic product placement or promotional bundling. Placing item ‘I’ physically next to ‘P’ on shelves or suggesting it in an online checkout flow will organically drive up the average basket size.

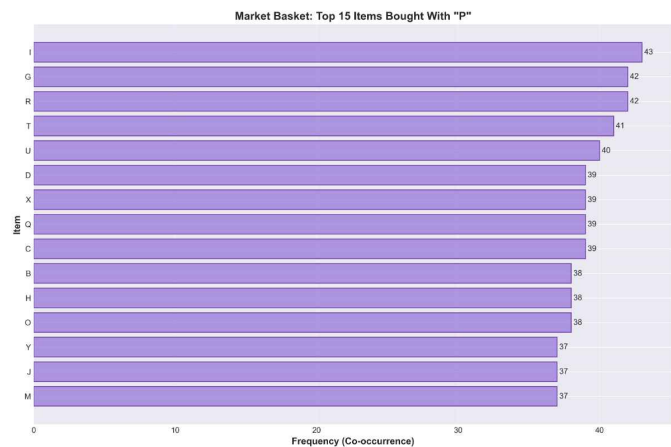


Fig. 8. Market Basket Analysis: Top Items Bought with Target Item ‘P’

Mapping the demand elasticity by plotting the average price of an item against its total transaction frequency reveals a parabolic trend. Cheaper items generally maintain high frequencies, whereas the most expensive items suffer drops in volume but drive massive revenue spikes. The curve indicates an optimal “sweet spot” for mid-priced items that balance both volume and margin. Identifying where each item sits on this curve is critical for deciding which products can sustain a price increase without losing buyer volume.



Fig. 9. Item Demand Elasticity: Price vs Popularity

Tracking the daily lifecycle and volatility of the top 5 revenue-generating items over the 30-day period highlights severe fluctuations in demand. High-revenue items experience extreme single-day spikes (e.g., Item ‘V’ nearing Rs. 4000 in a single day) followed by periods of near-zero sales, requiring robust, agile inventory management to prevent stockouts. This suggests that top-tier items are highly sensitive to specific buying periods and cannot rely on static daily restocking schedules.

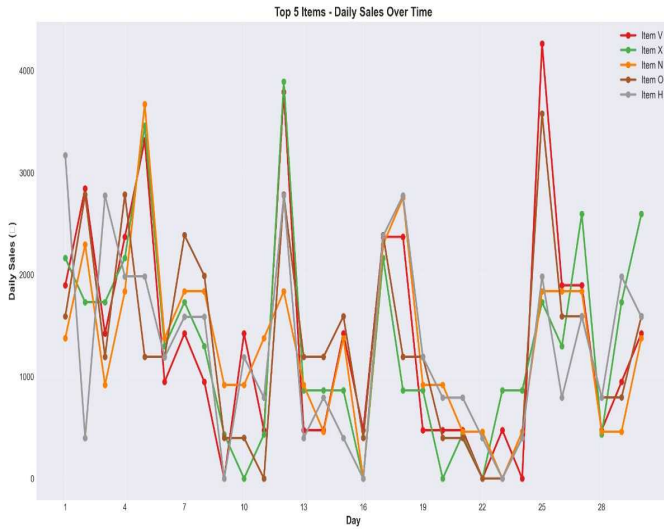


Fig. 10. Top 5 Items - Daily Sales Over Time

C. Transaction Analysis

The overarching metrics of the store's health rely heavily on temporal traffic analysis. A time-series plot of the daily transaction count uncovers significant cyclical demand, with traffic fluctuating drastically from a low of 2 transactions on Day 21 to peaks exceeding 18 transactions on multiple other days. These drastic drops and sharp climbs reveal that consumer foot traffic is highly clustered around specific, predictable windows rather than spread evenly across the month.

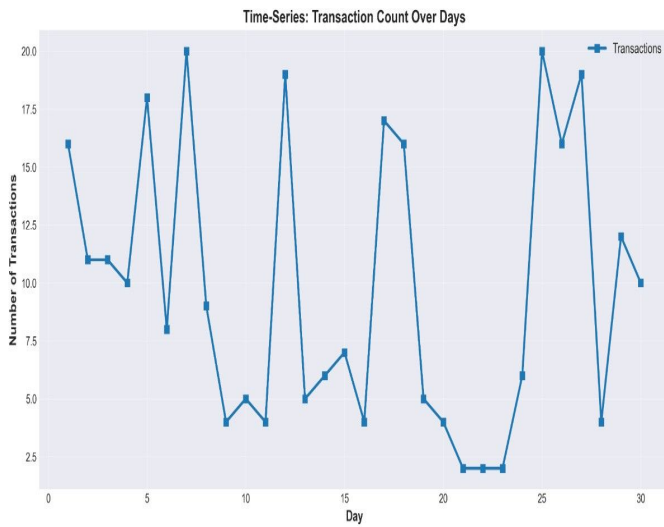


Fig. 11. Time-Series: Transaction Count Over Days

This cyclical operational demand is further clarified by explicitly highlighting the store's busiest days. A bar chart indicates that Days 7 and 25 represent the absolute peak operational load, both logging exactly 20 transactions. Recognizing these temporal spikes is crucial for optimal staff scheduling and promotional planning. Ensuring maximum operational

capacity on these proven peak days will directly reduce friction at checkout.

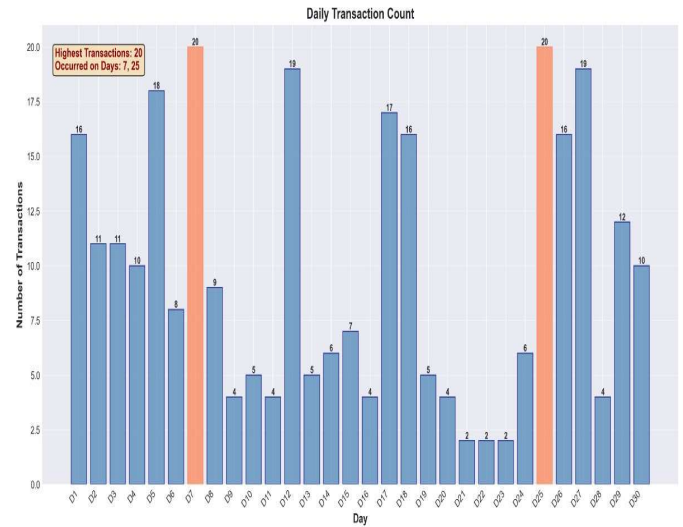


Fig. 12. Daily Transaction Count Highlighting Peak Activity Days

Assessing the relationship between the number of items in a user's cart and the final cost of that cart reveals an exceptionally strong positive correlation of 0.938. This strong linear trend confirms that adding items to the basket scales the transaction value reliably, indicating that large baskets are not artificially inflated by low-value filler items. Consequently, any business strategy focusing purely on increasing the total volume of items per basket is practically guaranteed to yield higher revenue.

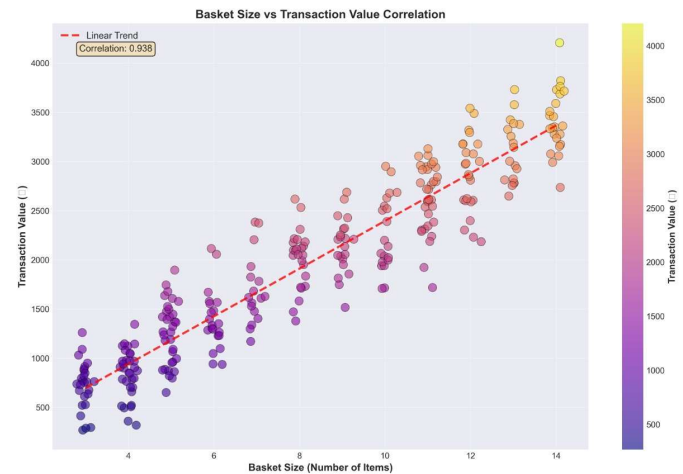


Fig. 13. Basket Size vs Transaction Value Correlation

Evaluating the concentration of wealth within the transaction dataset involves mapping the cumulative transaction percentages against cumulative revenue using a Lorenz Curve. The resulting Gini coefficient of 0.464 points to a moderate level of inequality, signifying that a relatively small percentage of high-value transactions contribute heavily to the overall

store revenue. The business relies heavily on these "whale" transactions to stay profitable, making large-basket shoppers extremely vital.

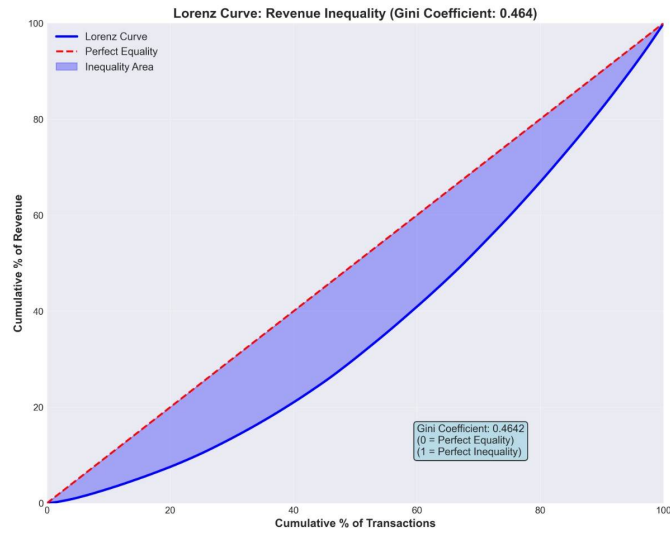


Fig. 14. Lorenz Curve: Revenue Inequality yielding a Gini Coefficient of 0.464

Finally, analyzing the total distribution of transaction sizes provides a macroscopic view of typical customer spending. The histogram displays a mean checkout value of Rs. 1966.65 and a median of Rs. 2002.05, accompanied by a slight right skewness of 0.114. The multi-modal spread suggests a highly diverse range of shopping trips, from minor top-ups to major hauls, confirming that the store caters to various levels of purchasing intent.

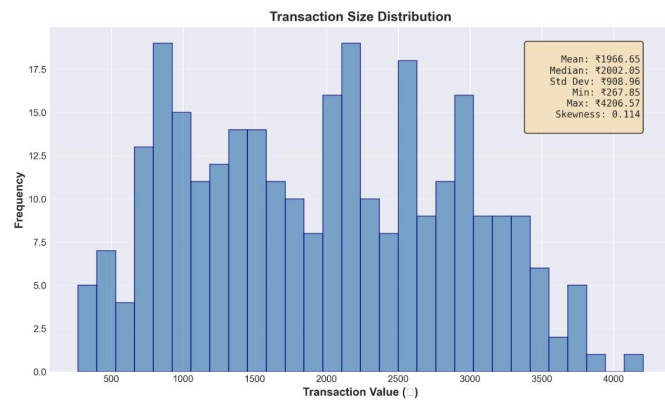


Fig. 15. Transaction Size Distribution and Summary Statistics

IV. CONCLUSION

This research demonstrates the efficacy of visualization and Exploratory Data Analysis methods in mapping the performance of a retail ecosystem. Utilizing dimensional parsing allows for precise estimation of user value and product lifecycle tracking without depending entirely on generic aggregations.

The findings emphasize that:

- 1) Specific users drive a disproportionate amount of the total Rs. 574,262 revenue, though high interaction frequency does not guarantee high total spend.
- 2) Identifying co-occurring items via Market Basket algorithms allows for highly targeted product bundling and cross-selling.
- 3) Revenue inequality (Gini: 0.464) and high basket-value correlation (0.938) signify a stable but moderately top-heavy transaction environment with clear operational peaks (Days 7 and 25).